Deep Learning Architectures for Gradient Artifact Reconstruction and Removal in Simultaneous EEG-fMRI: A Comprehensive Technical Report

1. Introduction: The Convergence of Electrophysiology and Hemodynamics

The integration of electroencephalography (EEG) and functional magnetic resonance imaging (fMRI) constitutes a methodological zenith in modern neuroimaging. This multimodal synthesis seeks to bridge the fundamental gap between the millisecond-scale electrophysiological dynamics of neuronal assemblies and the second-scale hemodynamic responses that support them.[1] For conditions characterized by paroxysmal electrical events, such as drug-resistant epilepsy, this concurrent acquisition is not merely advantageous but clinically imperative. It allows for the precise localization of the irritative zone by correlating interictal epileptiform discharges (IEDs)—transient spikes of 20 to 70 milliseconds duration [1]—with the Blood Oxygen Level Dependent (BOLD) signal measured by fMRI.

However, the simultaneous recording environment is electromagnetically hostile. The EEG system, designed to detect microvolt-level potentials (10^{-6} V) on the scalp, is placed inside the bore of an MRI scanner where it is subjected to static magnetic fields (B_0) of 1.5 to 7 Tesla, rapidly switching gradient fields (G_x, G_y, G_z) used for spatial encoding, and high-frequency radiofrequency (RF) pulses.[2]

The interaction between these external fields and the conductive loops formed by the EEG electrodes and the subject's scalp generates artifacts of immense magnitude. The **Gradient Artifact (GA)**, strictly linked to the MR image acquisition sequence, is the most dominant, often exceeding the amplitude of the biological EEG signal by a factor of 100 to 1,000.[2] While the EEG signal rarely exceeds 100 μV, gradient artifacts can reach tens of millivolts, effectively burying the neural signal of interest.[3]

This report provides an exhaustive analysis of the signal processing challenges posed by GA and explores the burgeoning field of Deep Learning (DL) solutions. Moving beyond traditional subtraction methods, we investigate advanced architectures ranging from Denoising Autoencoders and Generative Adversarial Networks to cutting-edge Diffusion Probabilistic Models and State-Space Models (Mamba). Furthermore, we propose novel, creative applications of architectures borrowed from audio source separation and computer vision,

theorizing their efficacy in the specific domain of EEG-fMRI artifact removal.

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2. The Physics of the Gradient Artifact: A Signal Processing Perspective

To design effective deep learning architectures, one must first understand the generative physics of the noise. The Gradient Artifact is not stochastic noise; it is a deterministic signal governed by Faraday's Law of Induction.

2.1 Electromagnetic Induction and Morphology

The fundamental equation governing the generation of the GA is:

$$\varepsilon = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}\int \mathbf{B} \cdot d\mathbf{A}$$

Where ε is the induced electromotive force (EMF), $\mathbf{B}$ is the magnetic field vector, and $\mathbf{A}$ is the area vector of the loop formed by the EEG leads and the scalp. In an fMRI sequence, specifically Echo Planar Imaging (EPI), the magnetic field gradients switch rapidly to encode spatial information.

- **Slice Selection Gradient:** Applied along the z-axis (typically) to select a slice.
- **Phase Encoding Gradient:** Applied prior to readout to encode the y-axis spatial information.
- **Frequency Encoding (Readout) Gradient:** Applied during signal acquisition to encode the x-axis.

Each of these switching events creates a massive dB/dt (change in magnetic flux over time). Since the EEG leads form a fixed loop on the head, this flux change induces a voltage spike. The "Gradient Artifact" is the superposition of the voltages induced by all three gradient coils.[2]

Because the MRI sequence is precisely timed, the GA is theoretically periodic. It repeats with every repetition of the slice acquisition (Slice Repetition Time) or volume acquisition (Volume Repetition Time, TR). This periodicity is the foundation of the traditional **Average Artifact Subtraction (AAS)** method.[2]

2.2 The Stationarity Problem

If the subject were perfectly immobilized and the scanner electronics thermally stable, AAS would be sufficient. It works by averaging the artifact over several windows to create a template, which is then subtracted from the current window.

However, clinical reality defies stationarity:

1. **Head Motion:** Even sub-millimeter movements of the patient's head alter the geometry of the induction loop (the angle between the loop area vector $\mathbf{A}$ and the field vector

$\mathbf{B}$). This changes the amplitude and shape of the induced artifact significantly.[3]

2. **Vibration:** The gradient coils themselves vibrate due to Lorentz forces ("scanner noise"), transmitting mechanical vibration to the EEG cap.
3. **Clock Drift:** Slight desynchronization between the EEG amplifier clock and the MRI sequencer clock leads to "jitter," where the sampling of the sharp gradient spikes varies, creating aliasing artifacts.[2]

The failure of AAS in the presence of motion leaves residual artifacts. These residuals are particularly pernicious because their spectral content—broadband spikes—overlaps with the gamma band (> 30 Hz) and the specific frequency profile of epileptiform spikes, leading to false positives in epilepsy diagnosis.[1]

2.3 The Ballistocardiogram (BCG) Confounder

While this report focuses on the Gradient Artifact, the BCG artifact is inextricably linked. The BCG arises from the pulsatile motion of the scalp caused by cardiac ejection. The blood rushing into the head causes a microscopic "nodding" motion of the head in the static B_0 field.

$$\varepsilon_{BCG} \propto v \times B_0$$

This motion (v) in the strong field generates the BCG. Crucially, methods that remove GA often distort the BCG or fail to account for the interaction between the two. A robust Deep Learning model must either implicitly handle both or be robust enough to remove GA without altering the BCG morphology, which is removed in a subsequent step.[1]

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3. The Discriminative Era: Autoencoders and CNNs

The first wave of deep learning solutions for EEG-fMRI artifact removal treated the problem as a supervised regression task: mapping a noisy input X to a clean target Y. The dominant architecture in this paradigm is the **Denoising Autoencoder (DAE)**.

3.1 Denoising Autoencoder (DAR) Framework

The **Denoising Autoencoder (DAR)** [1] represents a direct non-linear extension of template subtraction. Instead of a fixed template, the network learns a dynamic mapping function.

3.1.1 Architecture Specifics

The typical DAR for EEG uses a **1D Convolutional Neural Network (1D-CNN)**.

- **Input:** A segment of EEG (e.g., 2 seconds) sampled at high frequency (often downsampled to 250-500 Hz for processing, though GA requires higher rates).
- **Encoder:** A series of convolutional layers.
    - *Kernel Size:* The first layer often uses a larger kernel size (e.g., 5-10 ms) to capture

the broad shape of the gradient pulse.
    - *Stride:* Strided convolutions perform learnable downsampling, reducing the temporal dimension while increasing the channel depth (feature richness). This forces the network to compress the signal, discarding "incoherent" noise.
- **Bottleneck (Latent Space):** The highly compressed representation where the distinction between "signal" (neural) and "noise" (gradient) is maximized.
- **Decoder:** Transposed convolutional layers (deconvolution) upsample the latent features back to the original temporal resolution.
- **Skip Connections:** Many modern implementations utilize **U-Net** style skip connections.[7] These are critical for EEG because deep downsampling can destroy phase information. The skip connections allow high-frequency spatial/temporal details from the encoder to bypass the bottleneck and refine the output reconstruction.

3.1.2 Performance and Insights

Studies using the CWL EEG-fMRI dataset show that DAR achieves a Root Mean Squared Error (RMSE) of 0.0218 ± 0.0152 and a Structural Similarity Index (SSIM) of 0.8885, significantly outperforming PCA and ICA.[1]

Theoretical Insight: Why does a CNN work better than AAS? AAS assumes the artifact template is *additive* and *constant*. A CNN, through its non-linear activation functions (ReLU, Leaky ReLU), can model the artifact as a *function of the input state*. It learns that "when the signal rises sharply with slope m, it is likely a gradient switch," regardless of the exact amplitude, making it robust to the amplitude variations caused by head motion.

3.2 Dual-Pathway Autoencoders (DPAE)

A limitation of standard CNNs is the fixed receptive field. A small kernel sees sharp spikes but misses slow drifts (BCG). A large kernel sees drifts but blurs spikes. The **Dual-Pathway Autoencoder (DPAE)** [10] addresses this by splitting the encoder.

- **Local Pathway:** Uses small kernels and dilation to focus on high-frequency details (spikes, gamma activity).
- **Global Pathway:** Uses large kernels or pooling to capture low-frequency trends (BCG, slow cortical potentials).
- **Fusion:** The latent vectors from both pathways are concatenated. This ensures that the reconstruction (Decoder) has access to both the "texture" (neural oscillations) and the "structure" (artifact periodicity).

3.3 The U-Net Variation: IC-U-Net

IC-U-Net [8] introduces a domain-specific innovation: integrating **Independent Component Analysis (ICA)** into the training loop.

- **Concept:** Instead of training on raw channel data alone, the inputs are mixtures of Independent Components (ICs). The network learns to map these mixtures back to "clean" components.

- **Significance:** This bridges the gap between classical statistical signal processing and deep learning. ICA is excellent at identifying stationary sources; U-Net is excellent at non-linear reconstruction. By combining them, IC-U-Net leverages the statistical independence prior of ICA to guide the supervised learning of the CNN.

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4. The Generative Era: GANs and Adversarial Training

While Autoencoders minimize MSE (L2 loss), this often leads to "regression to the mean," resulting in smoothed signals where high-frequency textures (like the fine structure of an epileptic spike) are lost. **Generative Adversarial Networks (GANs)** solve this by introducing a critic.

4.1 The Adversarial Game for EEG

In the GAN framework [14], two networks compete:

1. **Generator (G):** Usually a U-Net or ResNet that outputs a "clean" EEG segment.
2. **Discriminator (D):** A classifier that tries to distinguish between the Generator's output and real, clean EEG (ground truth).

The loss function is a combination of reconstruction loss (L1/L2) and adversarial loss:

$$\mathcal{L} = \lambda_{rec}||G(x) - y||_1 + \lambda_{adv}(\log D(y) + \log(1 - D(G(x))))$$

Impact on Gradient Artifacts: The Discriminator learns the *distribution* of real EEG. If the Generator leaves a residual gradient spike, the Discriminator (recognizing that real EEG rarely has such steep slopes) will classify it as "Fake." This forces the Generator to be extremely aggressive in removing the specific morphology of the artifact, often achieving cleaner results than MSE-based methods which might just lower the amplitude of the spike.[14]

4.2 DHCT-GAN: Integrating Transformers

The **Dual-Branch Hybrid CNN-Transformer GAN (DHCT-GAN)** [15] represents the current apex of GAN architectures for this task.

- **Hybrid Encoder:** It combines CNN blocks (for local feature extraction) with **Transformer** blocks.
- **The Transformer Role:** Transformers use **Self-Attention** to model long-range dependencies. In the context of GA, the artifact repeats every TR (e.g., every 2 seconds). A standard CNN might not "remember" the artifact from 2 seconds ago. The Transformer's attention mechanism can attend to the previous occurrence of the artifact, using it as a reference to subtract the current one.
- **Dual-Branch Generator:** Specifically splits the latent space into "Clean Features" and "Artifact Features," forcing the network to explicitly model the noise rather than just treating it as a residual.[15]

4.3 Nested GANs

Nested GANs [17] tackle the problem in complementary domains.

- **Inner GAN:** Operates in the **Time-Frequency Domain** (Spectrogram). It is tasked with removing the specific spectral harmonics of the GA.
- **Outer GAN:** Operates in the **Time Domain**. It takes the output of the inner GAN and refines the waveform to ensure phase continuity.
  This "Divide and Conquer" strategy is highly effective because GA is sparse in the frequency domain (specific harmonics) but dense in the time domain.

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5. The Probabilistic Era: Diffusion Models

Diffusion Probabilistic Models (DPMs) are rapidly replacing GANs in image generation due to their training stability and mode coverage. In 2024-2025, they have emerged as the state-of-the-art for EEG artifact removal.[18]

5.1 The D4PM Architecture: Dual-Branch Diffusion

D4PM (Dual-branch Driven Denoising Diffusion Probabilistic Model) [18] is designed specifically for the blind source separation nature of EEG-fMRI.

5.1.1 Mechanism

Unlike GANs which map $Noise \rightarrow Data$ in one step, Diffusion models define a forward process that adds noise and a reverse process that removes it.

- **Forward Process:** $q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t I)$. Gradually turns EEG into Gaussian noise.
- **Reverse Process:** $p_\theta(x_{t-1}|x_t)$. A neural network predicts the noise added at each step to reverse the flow.

D4PM Innovation: Joint Posterior Sampling

D4PM does not just diffuse the noisy signal. It employs two coupled diffusion processes:

1. **Signal Branch:** Models the distribution of clean EEG (x_s).
2. **Artifact Branch:** Models the distribution of the gradient artifact (x_a).

During inference (sampling), the model uses a **Joint Posterior Sampler**.[20] At each step of the reverse diffusion, it estimates a candidate signal $\hat{x}_s$ and a candidate artifact $\hat{x}_a$. It then applies a **Consistency Constraint**:

$$\hat{x}_s + \hat{x}_a \approx x_{observed}$$

The model updates the gradients (scores) of both branches to satisfy this sum.

5.1.2 Why Diffusion Wins

This approach is technically superior because it treats GA removal as a **generative separation problem**. It acknowledges that the "clean" EEG could be one of many plausible signals and uses the "artifact" model to constrain the search space. By explicitly modeling the complex, non-linear manifold of the Gradient Artifact (which accounts for motion and vibration), the Artifact Branch acts as a "super-AAS" template generator that adapts to every millisecond of data.[20]
Results on benchmark datasets show D4PM achieves correlation coefficients (CC) exceeding **0.99**, surpassing essentially all GAN and Autoencoder baselines.[20]

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6. The Sequence Modeling Era: Mamba and State-Space Models

A critical constraint in GA removal is the **Sampling Rate**. To capture the steep dB/dt edges of the gradient pulses without aliasing, EEG is typically acquired at 5 kHz or higher. A 2-second epoch (one TR) thus contains 10,000 time points. Transformers ($O(N^2)$ complexity) choke on such long sequences. **Mamba (State Space Models)** offers a solution with **linear complexity** $O(N)$.[21]

6.1 Mamba and Structured State Space Duality (SSD)

Mamba is based on continuous-time state-space equations:

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t)$$

$$y(t) = \mathbf{C}h(t)$$

The innovation is the **Structured State Matrix ($\mathbf{A}$)** and the **Selective Scan Mechanism**.

- **HiPPO Matrix:** The matrix $\mathbf{A}$ is initialized using HiPPO theory (History Preserving Padé Operators), which mathematically guarantees the state $h(t)$ compresses the history of the input optimally.[23]
- **Selectivity:** The parameters $\mathbf{B}, \mathbf{C}, \Delta$ are input-dependent. The model can "choose" to reset its state or ignore inputs.
    - *Application to GA:* When a massive gradient spike hits, the model can effectively "gate" the input, preventing the artifact from corrupting the hidden state memory of the neural EEG. Conversely, it can "select" to remember the timing of the spike to predict the next one.

6.2 DenoiseMamba: The ConvSSD Module

DenoiseMamba [24] integrates this theory into a practical denoising architecture.

- **ConvSSD Module:** Combines a local 1D Convolution (to handle the immediate sharp edges of the artifact) with a Mamba/SSD layer (to handle the long-range periodicity).
- **Performance:** It achieves superior reconstruction accuracy compared to Transformers while requiring a fraction of the memory and compute, making it viable for high-sampling-rate processing.[26]

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7. Creative Cross-Pollination: Adapting Models from Audio and Vision

The user requested "creative" models not necessarily used for this case. This section proposes adapting architectures from **Audio Source Separation** and **Computer Vision**, theorizing their rigorous application to EEG-fMRI.

7.1 The "Cocktail Party" Analogy: Audio Source Separation

The problem of separating EEG from GA is mathematically isomorphic to the "Cocktail Party Problem" (separating a speaker from background noise). The GA is simply a very loud, rhythmic "speaker."

7.1.1 Conv-TasNet (Time-Domain Audio Separation Network)

Conv-TasNet [27] revolutionized speech separation by working directly on the waveform, bypassing the magnitude spectrogram (which loses phase information).

- **Relevance:** EEG phase is critical for connectivity analysis. Spectrogram-based denoising often distorts phase.
- **Architecture:** It uses a **Temporal Convolutional Network (TCN)** with stacked dilated convolutions. This creates an exponentially growing receptive field.
- **Proposal:** Train a Conv-TasNet where Source 1 is "EEG" and Source 2 is "Gradient." The model's ability to separate overlapping non-stationary sources in the time domain makes it ideally suited for separating the non-stationary GA (due to head motion) from the non-stationary EEG.

7.1.2 Demucs (Deep Music Separation)

Demucs [29] is a U-Net architecture with LSTM layers used to separate drums, bass, and vocals in music.

- **Relevance:** The Gradient Artifact is highly rhythmic, resembling a percussion track (drums). The EEG is more stochastic/harmonic, resembling vocals or strings.
- **Proposal:** Adapt Demucs to treat GA as the "drum track." Demucs is designed to handle the complex harmonics of musical instruments; the GA has a rich harmonic structure

(peaks at $1/TR, 2/TR, \cdots$). Demucs could theoretically model this harmonic series far better than a simple filter.

7.1.3 SepFormer (Separation Transformer)

SepFormer [31] uses a dual-path attention mechanism.

- **Intra-Chunk Attention:** Models local features (within a short window).
- **Inter-Chunk Attention:** Models dependencies between windows.
- **Proposal:** This is perfect for the **Slice vs. Volume** structure of fMRI. Intra-chunk can model the artifact within one slice acquisition. Inter-chunk can model the variation of the artifact across the entire volume (TR), capturing the slight modulations caused by the specific gradient axis active at that moment.

7.2 Computer Vision Adaptations: Spectrogram Inpainting

7.2.1 Vision Transformers (ViT) for Spectrograms

Instead of 1D time-series, convert the EEG to a **Time-Frequency Spectrogram**. The GA appears as distinct vertical lines (broadband spikes) and horizontal lines (harmonic frequencies).

- **Proposal:** Use a **Masked Autoencoder (MAE)** or **Vision Transformer (ViT)**.[33] Mask the regions of the spectrogram dominated by GA (the vertical lines). Train the ViT to "inpaint" the missing data based on the context of the clean EEG frequencies.
- **Benefit:** This leverages the massive pre-trained power of vision models. The model learns the "texture" of clean EEG spectrograms and hallucinates the missing data in the artifact-corrupted regions.

7.3 Graph Neural Networks (GNNs): Topographic Consistency

Current models mostly treat EEG channels independently or as a simple vector. They ignore the **3D geometry** of the head.

- **Concept:** The GA creates a specific electromagnetic field distribution inside the scanner bore. The artifact amplitude varies systematically across the scalp (stronger at the periphery/temples, weaker at the vertex/Cz).
- **Proposal:** Use a **Spatiotemporal Graph Neural Network (ST-GNN)**.[34] Define a graph where nodes are electrodes and edges represent spatial adjacency.
- **Mechanism:** The GNN can learn the **spatial signature** of the GA. If a massive spike occurs at Fp1 (frontal) but not at Oz (occipital), and the GNN knows the gradient field physics implies a global effect, it can use the correlation structure to better distinguish the artifact. Conversely, if a spike is localized to the temporal lobe (epilepsy) and doesn't follow the global gradient field distribution, the GNN preserves it.

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8. Implementation, Evaluation, and Real-Time

Constraints

8.1 The Ground Truth Paradox and Self-Supervision

A major hurdle is training data. We never have truly "clean" EEG simultaneous with fMRI.

- **Simulation:** Most papers use "semi-synthetic" data: taking clean EEG recorded outside the scanner and mathematically adding a template-based GA.[1]
    - *Critique:* This assumes the additive model is correct, which deep learning is supposed to surpass.
- **Self-Supervision (Noise-to-Noise):** A "creative" solution is **Noise2Noise** training. If we have two different noisy realizations of the same signal (e.g., two adjacent TRs where the EEG is similar but the motion-induced artifact variation is different), we can train a network to predict one from the other. The network will learn the common signal (EEG) and average out the uncorrelated noise variations.

8.2 Evaluation Metrics

Standard metrics (RMSE) are insufficient. A model can achieve low RMSE by simply zeroing out the signal (removing the massive artifact) but also killing the EEG.

- **Recommended Metrics:**
    - **Signal-to-Noise Ratio (SNR):** Standard log-ratio of powers.
    - **Spectral Coherence:** Check if the alpha/beta bands are preserved.
    - **Spike Detection Rate:** Run a standard epilepsy spike detector on the cleaned data. Does the deep learning model preserve the spikes better than AAS?
    - **Downstream BOLD Correlation:** The ultimate test. Does the cleaned EEG regressor produce a meaningful activation map in the fMRI GLM analysis?

8.3 Real-Time Deployment (Neurofeedback)

For fMRI-neurofeedback, data must be cleaned in <2 seconds (one TR).

- **Latency:** Diffusion models (D4PM) are slow due to iterative sampling (e.g., 50-100 steps). They are best for offline analysis.
- **Real-Time Candidate: Mamba (DenoiseMamba)** or **Conv-TasNet**. Their single-pass, linear/convolutional nature allows inference in milliseconds on modern GPUs, enabling true real-time multimodal neurofeedback.

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9. Conclusion

The removal of Gradient Artifacts from simultaneous EEG-fMRI has transitioned from rigid, template-based engineering (AAS) to flexible, data-driven Deep Learning.
The **Denoising Autoencoder (DAR)** proved that CNNs could learn the artifact manifold. **GANs** improved the spectral fidelity of the reconstruction. Today, **Diffusion Models (D4PM)** offer the highest fidelity by treating the problem as rigorous source separation with joint

posterior constraints.
However, the future likely belongs to **State Space Models (Mamba)** and **Audio-Inspired Architectures (Conv-TasNet)**. Mamba solves the sequence length bottleneck inherent in high-sampling-rate EEG, while audio models provide the perfect inductive bias for separating "loud" rhythmic noise from "quiet" stochastic signals.
For the researcher aiming to advance this field, the most fertile ground lies not just in deeper networks, but in **hybrid architectures**: using GNNs to enforce spatial physical constraints, Mamba to handle high-frequency temporal dependencies, and Diffusion to generate the probabilistic manifold of the artifact.

Table 1: Comparative Summary of Deep Learning Architectures for GA Removal

Architecture Class	Specific Models	Primary Mechanism	Strengths	Weaknesses	Best Use Case
Discriminative	DAR, DPAE, IC-U-Net	1D-CNN Mapping ($X \rightarrow$)	Fast inference, interpretable saliency.	MSE loss causes blurring; struggles with non-stationary phase.	Real-time monitoring; simple setups.
Generative (GAN)	DHCT-GAN, Nested-GAN	Adversarial Training (G vs D)	Preserves high-freq texture; sharp spike reconstruction.	Training instability (mode collapse); difficult hyperparameter tuning.	Preserving spectral power in gamma bands.
Probabilistic	D4PM, EEGDfus	Dual-Branch Diffusion, Joint Posterior Sampling	**State-of-the-Art fidelity**; explicitly models artifact manifold.	Slow inference (iterative); high compute cost.	Offline, high-precision epilepsy diagnosis.
Sequence (SSM)	DenoiseMamba, EEGM2	Structured State Space (HiPPO, SSD)	**Linear complexity ($O(N)$;** handles long sequences (>10k points).	Newer architecture; fewer pre-trained backbones available.	High-sampling-rate data (5kHz+); long-context dependency.
Audio-Inspired	Conv-TasNet, SepFormer, Demucs	Time-Domain Source Separation	Excellent at separating rhythmic "loud" noise from	Requires adaptation of audio priors to EEG channels.	Scenarios with heavy head motion (non-stationary artifacts).

			stochastic signal.		
Graph (GNN)	MSGM, ST-GNN	Spatiotemporal Topology	Enforces spatial consistency based on head geometry.	Computationally heavy for dense arrays (128+ channels).	High-density EEG; verifying artifact topography.

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